

4. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \epsilon$

Code:

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, n;

    printf("Enter string: ");
    scanf("%s", s);

    n = strlen(s);

    for(i = 0; i < n / 2; i++) {
        if(s[i] != s[n - 1 - i]) {
            printf("String does not belong to the CFG");
            return 0;
        }
    }

    printf("String belongs to the CFG");
    return 0;
}
```

Output:

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, n;

    printf("Enter string: ");
    scanf("%s", s);

    n = strlen(s);

    for(i = 0; i < n / 2; i++) {
        if(s[i] != s[n - 1 - i]) {
            printf("String does not belong to the CFG");
            return 0;
        }
    }

    printf("String belongs to the CFG");
    return 0;
}
```

```
Enter string: 101
String belongs to the CFG

=== Code Execution Successful ===
```

